

Chapter 93 — Management of Upper Urinary Tract Obstruction

- **Obstruction may be intrinsic, extrinsic, congenital or iatrogenic;** in many patients the cause is not immediately evident.
- **Treatment spans ureteral stent placement to ileal interposition;** endourology, laparoscopy and robotics are prominent, so the urologist must understand the technical considerations and risks of every option.
- The chapter is **organised by anatomical level of obstruction:** ureteropelvic junction, retrocaval ureter, ureteral stricture, ureteroenteric anastomosis, retroperitoneal fibrosis.

PATIENT PRESENTATION AND DIAGNOSTIC STUDIES

- **Ureteropelvic junction obstruction (UPJO) is usually congenital but can present at any age.**
- **Neonates and infants:** historically a **palpable flank mass**; now most often **asymptomatic hydronephrosis on maternal prenatal ultrasonography**, a large share of which proves to be UPJO.
- **Other routes to diagnosis:** **azotaemia** (bilateral obstruction or obstruction of a functionally or anatomically solitary kidney); incidental finding while imaging unrelated anomalies (e.g. congenital heart disease).
- **Older children and adults:** **intermittent abdominal or flank pain**, often with nausea or vomiting; **haematuria** (spontaneous or after minor trauma); microhaematuria, pyuria or frank urinary tract infection (UTI); **rarely hypertension**.

Imaging

- **Goal of imaging:** **anatomical location and functional significance** of an apparent obstruction.
- **Diuretic renography gives an accurate diagnosis in most patients:** patient well hydrated → **furosemide 0.3–0.5 mg/kg** → **diuretic given 20 minutes into the study**, allowing the collecting system to fill.
- **Technetium-99m mercaptoacetyltriglycine (^{99m}Tc-MAG3) is the preferred isotope** over ^{99m}Tc-diethylenetriaminepentaacetic acid (DTPA) or radio-iodinated Hippuran, because of imaging quality and dosimetry.
- Renography quantifies **differential renal function and degree of obstruction even in hydronephrotic units**, and helps localise the level of obstruction; it is non-invasive, widely available, and best for serial follow-up when a **standard protocol** is used.
- **P40 (percentage of maximal tracer remaining at 40 minutes)** raises the sensitivity of the renal scan (from about half to about three-quarters).
- Renography helps **exclude concomitant UPJO in children with high-grade reflux; MAG3 renography is the most accurate study after therapeutic intervention**.
- **Obstructed MAG3 curve:** T½ (drainage half-time) **>40 minutes** despite furosemide; after successful pyeloplasty the kidney drains spontaneously, even before furosemide.

COMPARISON — IMAGING MODALITIES IN SUSPECTED UPJO

Modality	Role	Key point
Renal ultrasonography	First study in neonates and infants	Large medial sonolucent pelvis with smaller rounded calyces joined by dilated infundibula → distinguishes UPJO from multicystic kidney in most; ureter not seen. Prenatal pelvic dilation predicts later need for repair

Modality	Role	Key point
Non-contrast computed tomography (CT)	Standard first test for acute flank pain	Hydronephrosis to the UPJ without calculus suggests UPJO; excludes radiolucent calculi, urothelial tumour
Contrast-enhanced CT	Anatomy and function	Delayed pelvic filling and contrast hold-up; can substitute for renography when assessing for repair; shows crossing vessels
Diuretic renography (MAG3)	Functional standard	Differential function, T½; best follow-up study
Retrograde pyelography	Confirms site and nature of obstruction	Usually at the time of repair to avoid introducing infection; emergently when acute decompression needed
Percutaneous nephrostomy + antegrade study	When retrograde fails or is hazardous (neonates, infants)	Decompresses, defines anatomy, tests recoverability of function , permits Whitaker test

- **Best practice combines an anatomical and a functional study** (e.g. retrograde pyelogram + diuretic renogram) to plan therapy.

Whitaker Test (Pressure-Perfusion Study)

- **Sequence:** nephrostomy access → **bladder drained by catheter** (prevents transmission of intravesical pressure) → renal pelvis perfused at **10 mL/min** with saline or dilute contrast under fluoroscopy → pelvic pressure and trans-UPJ gradient measured.
- First described by **Whitaker in 1973**.

WHITAKER TEST INTERPRETATION

Non-obstructed	Obstructed
Pelvic pressure up to 12–15 cm H₂O	Pelvic pressure >15–20 cm H₂O

- **Equivocal results are uncommon** (well under one in twenty).
- **Sources of inaccuracy:** variation in pelvic anatomy and compliance; patient position.
- **Invasive → reserve for cases non-diagnostic on less invasive tests.** The final decision collates presentation and all studies.

Concept — *Dilation is not obstruction. Every test in this section tries to separate an anatomical finding from functional impairment. Anatomical imaging (ultrasonography, CT, retrograde) shows where the problem is, and functional testing (renography, Whitaker) shows whether it matters.*

Indications and Options for Intervention

- **Indications:** symptoms; impaired overall function or progressive loss of ipsilateral function; stones or infection; rarely causal hypertension.
- **Goals:** relieve symptoms and preserve renal function, traditionally by **reconstruction restoring unobstructed flow**.
- **Children:** early repair gives the best chance of functional recovery, but **timing in neonates is controversial** because kidneys truly at risk are hard to identify.
- **Selective non-operative management of neonatal hydronephrosis is supported:** in a prospective series only a small minority needed pyeloplasty. Surgery was triggered by **progressive hydronephrosis or a 10% fall in differential function**, and all treated kidneys recovered to baseline.
- **Adults:** UPJO may not appear until middle age; if asymptomatic and physiologically indeterminate → **observation with serial renography**. A minority come to surgery over several years, and some lose a small amount of function. **Most affected patients ultimately benefit from reconstruction.**

- **Procedure of choice:** historically **open dismembered pyeloplasty**. **Robotic pyeloplasty is now primary therapy at most major centres**. Endourological incision remains an alternative, particularly for **secondary UPJO**.
- **Utilisation trend:** over a nine-year period minimally invasive pyeloplasty rose about **tenfold**, open fell by about 40% and endopyelotomy was stable. A comparative-effectiveness study showed **higher failure with endopyelotomy**, with **open and minimally invasive pyeloplasty equivalent**.

DETERMINANTS OF ENDOPYELOTOMY SUCCESS (LANDMARK PROSPECTIVE SERIES)

Subgroup	Success
Overall	73%
Crossing vessel present vs absent	42% vs 86%
High-grade vs low-grade obstruction	60% vs 81%
Both adverse factors excluded	95% (comparable with open pyeloplasty)

- **What the table tests:** patient selection. Crossing vessels and severe hydronephrosis are the two factors that drag endopyelotomy below pyeloplasty.
- **Trap:** other series found these factors less important, so crossing vessels are a **prognostic factor, not a contraindication**.

Primary versus Secondary UPJO

- Because crossing vessels predict endourological failure, **primary UPJO is increasingly repaired open, laparoscopically or robotically**.
- **Secondary UPJO — cross over: failed endourology → pyeloplasty** (open, laparoscopic, robotic); **failed pyeloplasty → endourology**, which still succeeds often enough to warrant an attempt.
- **Robotic redo pyeloplasty** is technically demanding but gives excellent results with adjuncts such as **buccal mucosa graft** and **near-infrared fluorescence**.

Nephrectomy

- **Primary nephrectomy:** poorly functioning or non-functioning moiety + **normal contralateral kidney**, often with UTI or pain; imaging shows a thin parenchymal shell.
- **Adults: differential function <15% = non-salvageable.**
- **If salvageability is unclear → temporary stent or nephrostomy, then reassess function.**
- **Children:** pyeloplasty and nephrectomy carry similar morbidity for kidneys <20% function. One group recommends **nephrectomy below 10%**, while others report stable function after pyeloplasty in kidneys <15%. **No agreed cut-off.**
- **Other indications:** extensive stones with chronic infection and poor function; **multiple failed repairs**. Only if the **contralateral kidney is adequate**.

Concept — *The choice of intervention follows the anatomy. Pyeloplasty handles almost any configuration. Endourology depends on selection (degree of hydronephrosis, function, calculi, crossing vessels). Nephrectomy is reserved for the kidney not worth saving when a good kidney exists on the other side.*

Options for Intervention

Endourological Management — Principles

- **History:** Ramsay 1984 “percutaneous pyelolysis” → Badlani 1986 coined “**endopyelotomy**”. The concept derives from Davis (1943) “**intubated ureterotomy**.”

- **Core concept: full-thickness lateral incision** of the proximal ureter from lumen to peripelvic/periureteral fat → stent across → heal.
- **Retrograde ureteroscopic holmium laser incision under direct vision** is the commonest approach today. Efficacy differs little between techniques; they differ in technical detail and complications.
- **Pyeloplasty success ≈95%**: it gives a wide, dependent, funnelled UPJ and permits pelvic reduction.
- **Endourology offers a shorter stay and quicker recovery, but lower success.**

Percutaneous Antegrade Endopyelotomy

Indications and contraindications.

- Indications as for any UPJO. Historically it was offered only with concomitant stones or after failed pyeloplasty, then briefly as primary therapy for most patients.
- **Remains appropriate for UPJO + pyelocalyceal stones needing percutaneous access**, since both are treated at one sitting.
- **Contraindications (any endourological approach): obstructed segment >2 cm, active infection, untreated coagulopathy.** Such cases can often be managed robotically.
- **Crossing vessels alone are not a contraindication**; marked **vessel entanglement** of the UPJ may defeat any endourological approach and is identified by **three-dimensional (3D) helical CT**.

Patient preparation.

- Work-up as for any renal operation; anaesthetic risk assessment; **sterile urine** at definitive intervention. If infection cannot clear because of obstruction → temporise with stent or nephrostomy.
- **Counselling**: success lower than formal reconstruction; bleeding needing transfusion, urinary leak, drain problems, **hydropneumothorax with upper-pole access**.

Technique.

- **No incision until the UPJ is crossed.** Retrograde: hydrophilic wire through an open-ended catheter → catheter advanced into pelvis → contrast to guide puncture.
- **Access**: prone; **mid-posterior or superolateral calyx**. **Not lower pole**, whose acute infundibular angle limits access across the UPJ.
- UPJ intubated antegrade under fluoroscopy, or a retrograde wire is grasped from above (through-and-through) → **working + safety wire**.
- **Incising tools**: cold knife (original), electrocautery, holmium:YAG (holmium:yttrium-aluminium-garnet) laser, balloon endodilators, and more recently the thulium fibre laser.
- **Incise laterally**: anatomical studies show the lateral aspect is **devoid of crossing vessels**. Direct vision lets any vessel be seen and avoided.
- **High insertion**: extend the incision to the dependent pelvis, bridging lateral ureteric wall to medial pelvic wall. Where the ureter inserts on the **anterior** pelvis, the **marcupialising incision must run truly posteriorly**.
- **Stent**: no consensus on size or duration. **14/7-Fr endopyelotomy stent** (wide end across UPJ); if not pre-stented and passage is difficult, **10/7-Fr or a standard 8-Fr stent** without compromising outcome. Evidence on large stents (up to 27 Fr) is conflicting.

Postoperative care and follow-up.

- **No strenuous activity for 8–10 days.**
- **Stent duration**: the traditional **6 weeks (Davis)** is still used; 2 versus 4 weeks showed no difference. Daily suppressive antibiotics are common but not evidence-based.

- **Follow-up: 1 month after stent removal** (history, examination, urinalysis, diuretic renogram) → **6 months** → **12-monthly**.
- Most failures occur in **year 1**, but late failures occur, so **follow adults for 2–3 years**: some failures still appear at 36 months, few by 60 months.

Results.

- Better than open pyeloplasty for pain, stay and return to activity. Retrograde endopyelotomy and laparoscopic/robotic pyeloplasty now offer similar convalescence.
- **Success**: pooled reviews ≈ **73.5% (range 57–100%)**; experienced centres **85–90%**, with **primary ≈ secondary UPJO**; long-term cold knife/holmium series about two-thirds; also feasible in transplant kidneys.

COMPARISON — LONG-TERM DURABILITY: ANTEGRADE ENDOPYELOTOMY VS PYELOPLASTY

Follow-up	Pyeloplasty	Endopyelotomy
3 years	85%	63%
5 years	80%	55%
10 years	75%	41%

- **What it tests**: both decline with time, but **the gap widens**. Endopyelotomy remains viable only in selected patients.
- **After failure**: retrograde or repeat percutaneous endopyelotomy, or laparoscopic/robotic/open pyeloplasty (outcomes not compromised, though operating time is longer). **CT angiography to exclude a crossing vessel**; if one is significant → **do not repeat endopyelotomy**.

Complications.

- **Analogous to percutaneous nephrolithotomy (PCNL)**, but **bleeding risk may be higher** because the UPJO parenchyma is thin and atrophic.
- **Bleeding — stepwise**: bed rest, hydration, transfusion if needed → **do not irrigate the nephrostomy** (let the system tamponade) → persistent bleeding → **selective angiographic embolisation**. Keep a **low threshold for angiography**, since successful embolisation usually avoids exploration and possible nephrectomy.
- **Infection**: sterilise urine first. The American Urological Association (AUA) recommends **first- or second-generation cephalosporin** prophylaxis, adjusted to local antibiograms. Consider prophylaxis while the stent is in (especially in women).
- **Early persistent obstruction is rare** (stent in place). Clot obstruction of the stent → a few more days of nephrostomy drainage.

Special considerations.

- **UPJO + stones: remove stones first, then incise**; otherwise fragments migrate into periureteral tissue and cause fibrosis or granuloma.
- **Pelvic stone**: the “UPJO” may be **stone-induced oedema** → clear stone, then reassess the UPJ; a retained nephrostomy makes a **Whitaker test** straightforward. **Solitary lower-pole stones pose no such dilemma** → combined percutaneous treatment.
- Laparoscopic/robotic pyeloplasty with stone removal is equally effective; the choice rests on **stone burden and surgeon experience**.
- **Percutaneous endopyeloplasty**: uncommon hybrid, an **endoscopic Heineke–Mikulicz repair** through a percutaneous tract. Good short-term results, feasible with laparoscopic needle-holders and a nephroscope; **may fail in secondary UPJO** because scar limits endoscopic reconstruction.

Retrograde Ureteroscopic Endopyelotomy

- **History:** combined percutaneous and flexible ureteroscopic management of an obliterated UPJ (1985) → ureteroscopic “pyelolysis” (1986) → cutting electrode through large ureteroscopes, still with a nephrostomy (a combined approach). Good early success, but **distal ureteral strictures from large rigid instruments**.
- **Now a single-stage retrograde procedure and the standard.** Advantage: **direct vision of a correctly sited full-thickness incision without percutaneous access**.
- **Cost:** retrograde ureteroscopic endopyelotomy is the most cost-effective when failures are counted. Endopyelotomy is cheaper than pyeloplasty, while open and minimally invasive pyeloplasty are cost-neutral, so **minimally invasive pyeloplasty is the best option where available**.
- **Indication:** functionally significant obstruction.
- **Contraindications:** **long obstruction; upper-tract stones**, better treated simultaneously by another route. With marked hydronephrosis the antegrade route may be more effective.

Technique.

- **Instrument:** women, sometimes a **6.9-Fr semirigid** ureteroscope; usually, and in men, **flexible ureteroscope via access sheath**. **General anaesthesia**, to prevent movement.
- **Sequence:** retrograde pyelogram → hydrophilic wire coiled in collecting system → **access sheath with safety wire across UPJ** → flexible ureteroscope to UPJ.
- **Narrow ureter:** dilate the intramural ureter (5-mm balloon or 9–10-Fr introducer). If still narrow → **stent and defer 5–10 days** for passive dilation.
- **Incision:** drain the pelvis → **200-µm holmium fibre** (200 or 365 µm through semirigid scope) → **0.8–1.2 J, 10–15 Hz** → incise **laterally** while withdrawing across the UPJ → repeat, deepening into peripelvic/periueteral fat → extend cranially and caudally into normal ureter until widely patent.
- Contrast extravasation confirms depth but is **usually unnecessary under direct vision**. Optional **balloon dilation to 24 Fr** completes the incision.
- **Bleeding:** haemostatic laser settings, or **reinflate the balloon for 10 minutes** of tamponade.
- **Balloon dilation alone** (36 Fr creates linear UPJ splits experimentally) → **success falls with time, to about 42%**.
- **Finish:** stent over wire → **Foley catheter** (prevents reflux/extravasation, reveals bleeding) → **renogram 4 weeks after stent removal** → review every 6–12 months for **24–36 months**.

Results.

- Holmium series: **70–80% success out to 5 years**; larger series 60–73%. A literature review gives **44–89%**.
- **Excluding crossing vessels >4 mm** on preoperative ultrasonography raises success to about 90%. Endoluminal ultrasonography can detect vessels but is rarely used, since less invasive imaging gives the same data.
- **Best retrograde results still lag behind pyeloplasty.**

Complications.

- Fewer with small instruments and fibres: **ureteral stricture rare**, embolisation and nephrectomy rare. Mostly **urinary leak, stent migration, infection**.
- **Ureteroarterial fistula** has been reported 2 weeks after laser endopyelotomy (managed ureteroscopically).

Other Retrograde Techniques

- **Fluoroscopically guided cautery wire balloon** (first clinical series 1993): success equal to ureteroscopic endopyelotomy but **more major complications (transfusion, embolisation) and more haemorrhage**.
- → **Ureteroscopic endopyelotomy is the retrograde approach of choice.**

COMPARISON — ENDOUROLOGICAL OPTIONS FOR UPJO

Technique	Best use	Key advantage	Key drawback
Percutaneous antegrade endopyelotomy	UPJO + renal stones; marked hydronephrosis	Stones treated at same sitting; direct vision	PCNL-type bleeding (thin parenchyma); hydropneumothorax (upper pole)
Retrograde ureteroscopic laser endopyelotomy	Standard endourological route	Direct vision without a tract; most cost-effective	Contraindicated with stones or long segments
Cautery wire balloon	Historical	Fluoroscopic simplicity	More haemorrhage, transfusion, embolisation (blind)
Balloon dilation alone	Not recommended	Simple	Success decays to about 42%
Percutaneous endopyeloplasty	Selected primary UPJO	Heineke–Mikulicz reconstruction	Fails with secondary scar

KEY POINTS: ENDOUROLOGICAL MANAGEMENT OF UPJO

- **Indications:** symptoms, overall or progressive ipsilateral functional loss, stones or infection, rarely causal hypertension.
- **Endoscopy avoids the abdomen but does not match pyeloplasty success.**
- **Incise under direct vision.**
- **Low threshold for angiography** after bleeding from percutaneous endopyelotomy; embolisation obviates exploration (which risks nephrectomy).

Concept — Endopyelotomy works by **controlled full-thickness incision and healing over a stent**. It fails where healing cannot open the lumen: a **crossing vessel** that the incision cannot address, a **capacious non-draining pelvis**, or a **long segment**. Direct vision and a lateral incision exist to avoid the vessel you cannot see.

Reconstructive Interventions

Historical Notes

COMPARISON — EVOLUTION OF UPJ RECONSTRUCTION

Year	Surgeon	Contribution	Distinguishing feature / flaw
1886	Trendelenburg	First reconstruction	Patient died
1891	Kuster	First successful dismembered pyeloplasty	Prone to resticture
1892	Fenger	Heineke–Mikulicz (longitudinal incision, transverse closure)	Shortens one side → buckling, kinking
1916 / 1937	Schwytzer / Foley	Y-V pyeloplasty	Only for high insertion ; useless if UPJ already dependent
1951	Culp–DeWeerd	Spiral flap	More universally applicable flap
1953	Scardino–Prince	Vertical flap	—
1969	Thompson	Renal capsular flap	When pelvis inadequate

Year	Surgeon	Contribution	Distinguishing feature / flaw
1949	Nesbit	Elliptical dismembered anastomosis	Reduces stricture
1949	Anderson-Hynes	Spatulated ureter to projection of lower pelvis after excising redundancy	Today's standard
1943	Davis	Intubated ureterotomy (earlier: Fiori, Albarran, Keyes)	Healing by secondary intention

- Minimally invasive reconstruction has risen dramatically, yet **open surgery remained common** (about 41% of pyeloplasties in a 2002–2010 claims analysis).
- **Universal principles: widely patent, watertight, tension-free anastomosis; funnel-shaped pelvis-to-ureter transition in a dependent position.** Minimally invasive surgery mimics open surgery, so the principles are shared.
- **Pre-repair drainage only in select cases: infection; azotaemia from a solitary kidney or bilateral disease; rarely severe, unrelenting pain.** Use a stent or nephrostomy.

Dismembered Pyeloplasty

Indications.

- **Preferred by most urologists: almost universally applicable.** Works with high or dependent insertion; allows **pelvic reduction**, straightening of a tortuous ureter and **anterior/posterior transposition over crossing lower-pole vessels**.
- **Only the dismembered technique excises the abnormal UPJ completely** (flaps do not).
- **Poorly suited to: long or multiple proximal ureteral strictures; small, inaccessible intrarenal pelvis.**

Technique (open and minimally invasive identical).

- **Sequence:** identify proximal ureter → dissect cephalad **with generous periureteral tissue** (blood supply) → **lateral marking stitch below the obstruction** for orientation → excise UPJ → **spatulate ureter laterally** → **apex of spatulation to inferior border of pelvis**, medial side to superior aspect → full-thickness, watertight anastomosis with fine interrupted or running absorbable suture.
- **Reduction pyeloplasty** for a very redundant pelvis (often unnecessary): excise redundancy → close cephalad pelvis down to the dependent part → anastomose.
- **Crossing vessels:** dismemberment allows the UPJ to be transposed to the other side of the vessel.
- **Stent:** most place a **soft, self-retaining internal stent, removed at 4–6 weeks** under local anaesthesia. It **reduces extravasation** and so secondary fibrosis, and allows earlier drain removal.
- **Nephrostomy + stent in uncomplicated adult repair gives no advantage** (longer stay, more infection). Reserve for **secondary UPJO or active inflammation**; keep a pre-existing nephrostomy for diversion and antegrade studies.
- **External drain:** closed suction **near but not on the suture line**, via a separate stab incision, to prevent urinoma, disruption, scarring or sepsis.

Surgical Approaches for Pyeloplasty

Open surgery.

COMPARISON — OPEN APPROACHES TO THE UPJ

Approach	Advantage	Best suited to
Extraperitoneal flank (subcostal; usually 12th-rib bed or off its tip)	Familiar; excellent exposure regardless of habitus	Most surgeons' default

Approach	Advantage	Best suited to
Anterior extraperitoneal	Minimal mobilisation	Thin patients; horseshoe or pelvic kidney (laparoscopy also possible)
Posterior lumbotomy	Direct UPJ exposure, minimal mobilisation	Thin patients without previous ipsilateral surgery

Laparoscopic and robotic intervention.

- **Laparoscopic pyeloplasty first 1993**; greater complexity and steeper learning curve than open or endopyelotomy. It gives less morbidity, shorter stay and faster convalescence with **success ≥90%**, **matching open**.
- **Meta-analysis**: open and minimally invasive have **similar success and complications**; stay about **2.7 days shorter** with minimally invasive. Pyeloplasty beats endopyelotomy by **about 10–30%**; large laparoscopic series report 94–100% (short follow-up).
- **Robot (first clinical case 2002)**: shorter learning curve, wristed suturing → **drove the rise in minimally invasive pyeloplasty**. US use rose from a few per cent (1998) to over half (2009), with robotics accounting for nearly half of all pyeloplasties by 2009.
- **Indications**: as for open/endourological repair. **Crossing-vessel transposition and pelvic reduction are both feasible**.
- **Absolute contraindications**: **uncorrected coagulopathy**; **untreated active UTI**; **cardiopulmonary compromise precluding surgery**.
- **Objective**: tension-free, watertight, funnel-shaped repair.
- Dismembered **Anderson-Hynes** is preferred laparoscopically too; Y-V and flap (Culp) methods are also possible.

COMPARISON — MINIMALLY INVASIVE ACCESS ROUTES FOR PYELOPLASTY

Route	Set-up	Advantage	Limitation / note
Transperitoneal (1993)	45° lateral decubitus; Veress or Hasson access; 3–5 ports, umbilical camera; medial colonic mobilisation	Most widely used : large working space, familiar anatomy	Colon must be reflected
Transmesenteric modification	Open mesocolon directly over UPJ	Shorter operation , possibly shorter stay (less bowel handling)	Needs a visible dilated pelvis: thin, young patients; pre-stenting deflates the pelvis and hides it
Retroperitoneal (1996)	Flank, kidney rest; Hasson entry; balloon-dilated space ; 3–4 ports	Ureter found early; stays extraperitoneal	Smaller space
Anterior extraperitoneal (2003)	Peritoneal sac and bowel mobilised medially en bloc	Wholly extraperitoneal with anterior view	Space created manually
Robotic-assisted (experimental 1999; clinical 2002)	Usually transperitoneal; 4 trocars (3 robotic incl. camera + 1 assistant)	3D vision, motion scaling, tremor filtration, dexterity, range of motion	Retroperitoneal feasible
Single-port (SP)	2.5–3-cm (peri)umbilical Hasson incision; port device or three 5-mm ports; 5-mm scope, bent/articulating instruments	Cosmesis ; no specimen to extract	Loss of triangulation , instrument clash, hard suturing (accessory needlescopic port sometimes)

Route	Set-up	Advantage	Limitation / note
Robotic SP (da Vinci SP, FDA-approved 2018)	Flexible scope + 3 wristed arms through one port	Restores triangulation; no major complications in early series	Operating time ≈50% longer than multiport; stay, blood loss, pain, 30-day complications similar

Transperitoneal technique (template for all routes).

- **Sequence:** optional cystoscopy, retrograde pyelogram, stent and Foley → position and pneumoperitoneum → mobilise colon medially (or transmesenteric window) → dissect ureter cephalad to UPJ and pelvis (**avoid extensive dissection and diathermy near the ureter**) → define cause → **transect pelvis above UPJ, spatulate ureter laterally** → transpose over any crossing vessel → intracorporeal anastomosis (**4-0 absorbable**, running or interrupted; freehand or EndoStitch) → optional reduction pelvioplasty → drain through a trocar site.
- **Antegrade stent option:** wire down the ureter via an upper-quadrant port or a **14-gauge angiocatheter** through the subcostal region → stent over wire. **Clamp the Foley** so the bladder fills: urine issuing from stent side-holes confirms the distal coil lies in the bladder.

Vascular transposition ("vascular hitch").

- Lower-pole crossing vessels are mobilised and **fixed cranially over the renal pelvis without dismembering the UPJ**.
- Used mainly in children, with high success (about 90–95%). **Pitfall:** uncertain whether the vessel is the **sole** cause of obstruction, since intrinsic UPJ disease is left untreated.

Postoperative care and complications.

- **Sequence:** clear fluids day 0 → antibiotic prophylaxis → **Foley out at 24–36 hours** → drain out before discharge if output negligible. **If drain output rises after Foley removal → replace Foley for 7 days** (reflux along stent) → **stent out at 4–6 weeks** → renography as after open repair.
- **Complications:** as for laparoscopic abdominal surgery: colonic injury, haemorrhage, ileus, pneumonia, heart failure, thrombophlebitis, **urinoma**. Roughly 2% intraoperative and 13–16% postoperative.

Results

- **Open dismembered pyeloplasty:** very durable, **almost never ends in nephrectomy**. It resolves symptoms in about 95% and decompresses the pelvis in about 91%, **arrests functional decline in almost every case** and improves function in most.
- **Laparoscopic (mostly Anderson-Hynes, primary):** operating time 2–4 hours (under 3.5 hours when experienced); complications 2–16%; **conversion 0–5.5%**; transfusion anecdotal; **stay ≈2.7 days vs 4.2 open; success 87–99%, mostly >95%**. Safe in infants under 1 year.
- **Failures:** mostly within 2 years, but **up to 30% occur later**. Salvage: **open ≈86%, endopyelotomy ≈70%**, and endoscopy handles most. **Paediatric: about 1 in 9 need a secondary procedure**, so failure is not rare.
- **Robotic:** operating time 100–299 minutes; complications 3–24%; conversion 0–6.8%; stay 2.2–2.8 days; **success 94.7–100%**, similar to laparoscopy; feasible in children.
- **Late failure matters:** in one long series success fell from **85% at 3 years to 75% at 10 years**. A robotic series reported about 92% failure-free survival at 8 years, but largely on telephone follow-up.
- **Renal anomalies** (horseshoe, pelvic): laparoscopic pyeloplasty is safe and effective.
- **Secondary UPJO:** laparoscopic repair after failed endopyelotomy succeeds in about 83%, with much longer operating time; after failed open repair success is high (about 89%).

TABLE 93.1 COMPARATIVE SERIES OF ROBOTIC VERSUS LAPAROSCOPIC PYELOPLASTY — DATA FROM SEVEN COMPARATIVE SERIES, 2006–2013

Endpoint	Laparoscopic pyeloplasty (LP)	Robotic-assisted pyeloplasty (RAP)	Pattern
Populations	Adults (mean age mid-20s to 40s); one paediatric-only series; one laparoendoscopic single-site (LESS) comparison		—
Operating time	About 80–270 min	About 100–300 min	No consistent winner. Only significant difference: LP shorter (80.7 ± 21.9 vs 100.2 ± 9.1 min, $P = 0.018$) in the earliest prospective series; RAP shorter in several later series
Estimated blood loss	40–100 mL	40–56 mL	Small either way; no significant difference
Hospital stay	0.9–5.5 days	1.5–2.8 days	RAP tends shorter in adult series
Complications	0–51.5%	0–23.5%	Wide spread; highest in one series for both arms
Success	88–100%	93–100%	Equivalent; “strict” success (no pain + no obstruction on nuclear scan) only about two-thirds in one series

- **What the table is for:** robotic pyeloplasty does not outperform laparoscopy on outcome. **Its advantage is ergonomic** (easier suturing, shorter learning curve), and that is why it spread.
- **Trap:** headline success depends on its definition. “Strict” criteria roughly one-third lower; short follow-up inflates success.

Special Situations of Laparoscopic and Robotic Management of UPJO

COMPARISON — MINIMALLY INVASIVE VARIANTS FOR SPECIAL ANATOMY

Procedure	Indication	Technical essence	Outcome
Ureterocalicostomy (laparoscopic/robotic)	UPJO + small pelvis and dilated lower-pole calyx	Pre-placed double-J stent; 45–60° flank; excise a rim of lower-pole parenchyma ; ligate pelvic opening; spatulated ureter to calyx, mucosa-to-mucosa	Largest (mostly redo) series about 70% success; thin parenchyma (cortex <5 mm) and poor function predict failure ; most failures in year 1; access route not a risk factor. Small series excellent (paediatric robotic: no obstruction at 12 months)
Pyeloplasty + pyelolithotomy	UPJO + calculi	Stones extracted through pyelotomy with graspers, flexible cystoscope/nephroscope via port (robotic: undock an arm)	Stone-free about 80–100%; UPJ patent ≈90%
Dismembered tubularised flap	Long upper-ureteral defect (e.g. 3 cm) after stricture excision	Wide-based pelvic flap tubularised to bridge gap	Patent at short follow-up
Calicovesicostomy	Low-lying obstructed moiety (e.g. horseshoe) + large bladder	Nephrotomy at most dependent lower pole → anastomosed to bladder dome	Patent at 4 months

KEY POINTS: LAPAROSCOPIC AND ROBOTIC INTERVENTION

- **Transperitoneal is the most widely used route** (large working space, familiar anatomy). Retroperitoneal and anterior extraperitoneal routes need a created space (manual or balloon).
- Laparoscopic UPJO repair gives **low complications, short stay and success >95%** in experienced hands.

Other Reconstructive Procedures (Non-Anderson-Hynes)

- Useful in specific anatomy; open or minimally invasive according to surgeon skill.

COMPARISON — FLAP AND ALTERNATIVE UPJ RECONSTRUCTIONS

Procedure	Specific indication	Contraindication / limitation	Geometry
Dismembered (Anderson-Hynes)	Almost any UPJO ; crossing vessels; redundant pelvis	Long/multiple proximal strictures; small intrarenal pelvis	Excise UPJ, spatulate, reanastomose
Foley Y-V-plasty	High ureteral insertion	Crossing vessels needing transposition ; pelvic reduction needed	V-flap base on dependent medial pelvis, apex at UPJ; stem of Y down lateral ureter into normal calibre
Culp-DeWeerd spiral flap	Large accessible extrarenal pelvis, dependent oblique UPJ , plus a long proximal ureteral narrowing	Length:width $\leq 3:1$	Oblique base lateral to UPJ (between insertion and parenchyma); spirals posterior $\leftrightarrow$ anterior; longest flap
Scardino-Prince vertical flap	Dependent UPJ at medial margin of a large box-shaped extrarenal pelvis	Limited use; shorter flap than spiral	Horizontal base; two straight converging incisions to apex
Davis intubated ureterotomy	Lengthy or multiple ureteral strictures , rarely used; combine with a flap (spiral best)	Relies on secondary healing	Distal ureterotomy left open to regenerate over stent; routine nephrostomy
Ureterocalicostomy	Small intrarenal pelvis ; rotational anomalies (horseshoe); salvage of failed pyeloplasty	Scar may prevent pelvic dissection	Resect lower-pole parenchyma (not simple nephrectomy)

Foley Y-V-plasty technique.

- **Sequence:** outline wide V-flap with methylene blue or stay sutures → ureterotomy through stenosis **several millimetres into normal ureter** → raise flap (scalpel, then Potts or fine Metzenbaum scissors) → stent → **flap apex to inferior apex of ureterotomy** → posterior walls → anterior walls.
- **Interrupted sutures preferred**, as they minimise pursing, buckling and ischaemia.

Spiral and vertical flap technique.

- **Flap length is set by the length of ureter to be bridged:** the farther the apex from the base, the longer the flap. **Length:width $\leq 3:1$** to keep it vascularised.
- **Outline longer than seems necessary** (the flap shrinks once the pelvis is opened); trim excess from the apex, never the base.
- Medial incision carried down the ureter through the stenosis → **rotate apex to lowest point of ureterotomy** → close over stent with fine absorbable suture.

Intubated ureterotomy technique.

- Flap outlined (spiral preferred: longer flap, **less ureter left to heal by secondary intention**). In this setting **any flap beats dismembered repair for blood supply**.
- Ureterotomy through the whole stricture(s); **minimal ureteral dissection**; **routine nephrostomy** (prevents urinoma, allows antegrade studies).
- Original: a stenting catheter exits through the cortex beside the nephrostomy. Now: **soft internal stent** (pelvis or lower infundibulum to bladder); flap apex brought as low as possible; distal ureterotomy left open, with a few loose sutures to keep the ureter against the stent.
- **At 6 weeks**: antegrade nephrostogram → no leak → stent out cystoscopically → repeat antegrade study → clamp → remove nephrostomy.

Ureterocalicostomy technique.

- **Sequence**: ureter dissected with generous periureteral tissue → kidney mobilised to expose lower pole → **resect lower-pole parenchyma** (partial lower-pole nephrectomy) to widely expose a dilated calyx → spatulate ureter laterally → stent ($\pm$ nephrostomy) → first stitch at spatulation apex to lateral calyx wall, second at 180° → **“open” interrupted technique**: all sutures placed, then tied → capsule closed over cut parenchyma but **not near the anastomosis** → cover anastomosis with **perinephric fat, peritoneal or omental flap** → urogram about 1 month after stent removal.
- **Why resect, not incise**: simple nephrotomy leads to secondary stricture.

Salvage Procedures

- **Failed open pyeloplasty → endourology first**. If unsuitable → redo flap or dismembered repair, aided by an **intraoperative ureteral catheter**; expect a long proximal stenosis needing **wide mobilisation of kidney and ureter** for a tension-free repair. Minimally invasive redo is feasible in skilled hands.
- **Other options**: ileal ureter; autotransplantation with Boari flap pyelovesicostomy; nephrectomy if the kidney is poor and the contralateral kidney normal.

Postoperative Care and Management of Complications

- **External drain out 24–48 hours after urinary drainage stops**; stent out at **4–6 weeks**.
- **Nephrostomy**: nephrostogram **no sooner than 7–10 days** (later if complex) → patent without leak → **clamp 12–24 hours** → remove if no pain, fever or leakage.

Concept — Every UPJ reconstruction answers one question: **what tissue bridges the gap while preserving blood supply?** Dismemberment excises the disease but devascularises. Flaps keep the blood supply but carry the disease forward. The calyx is used when the pelvis is too small. The choice between them is anatomical.

RETROCAVAL URETER

Aetiology and Diagnosis

- **Rare congenital anomaly from persistence of the posterior cardinal veins**; always right-sided.
- **Suspect on a characteristic S-shaped deformity** of the right ureter on intravenous or retrograde pyelography. The ureter courses **lateral to medial behind the inferior vena cava (IVC)**.
- **Definitive non-invasive diagnosis**: 3D CT.
- **Intervene only for functionally significant obstruction** causing pain or functional deterioration.

Operative Intervention

- **Standard repair: ureteroureterostomy (or ureteropelvic reanastomosis) with transposition anterior to the IVC.**
- **Sequence:** identify ureter, dilated pelvis and IVC → **transect dilated pelvis/ureter** → transpose ureter **anterior to the IVC** → circumferential tension-free, watertight absorbable anastomosis → drain + stent.

Laparoscopic and Robotic Surgical Management

- Successful in case series; transperitoneal (larger space) or retroperitoneal.
- **Sequence:** cystoscopic double-J stent → ≥4 ports (3 robotic + assistant) → mobilise ureter off IVC → **divide at the most distal part of the dilated segment** → excise redundant dilated and stenotic ureter → reposition anterolateral to IVC → **spatulate 1.5–2 cm on opposite sides** → intracorporeal anastomosis over stent → drain (out in days), stent out at 4–6 weeks.
- Minimal morbidity, short convalescence, patent on short-term imaging.

KEY POINTS: RETROCAVAL URETER

- **Persistent posterior cardinal veins; always right-sided.**
- Diagnosed by retrograde pyelography or CT.
- Intervene for functionally significant obstruction; open, laparoscopic and robotic repair all succeed.

Concept — *The anomaly is venous, not ureteric. The repair therefore moves the ureter, not the cava: divide it, bring it in front of the IVC, and rejoin it.*

URETERAL STRICTURE DISEASE

Aetiology

BOX 93.1 CAUSE OF URETERAL STRICTURE

- **Malignancy** (transitional cell carcinoma; cervical cancer)
- **Ureteral calculus**
- Radiation
- **Ischaemia or trauma from surgical dissection**
- Periureteral fibrosis (abdominal aortic aneurysm, endometriosis)
- **Endoscopic instrumentation**
- Renal ablation injury
- Infection (tuberculosis; also schistosomiasis)
- Idiopathic

- **What the box is for:** the list mixes intrinsic, extrinsic, iatrogenic and malignant causes. **Malignancy heads it because it must be excluded first.**
- **Trap:** urothelial carcinoma (**radiolucent filling defect, goblet sign**) can mimic a benign stricture. **Metastases** (cervix, prostate, ovary, breast, colon) can present as a stricture. So an **“idiopathic” stricture needs CT** to exclude intrinsic tumour or extrinsic compression.

- **Stones and their treatment are the key risk factors.** Stricture follows about 3% of ureteroscopies (URS) versus about 1.5% of shock-wave lithotripsy (SWL, taken to represent stone disease alone), roughly **1.7-fold higher after URS**.
- **Thulium laser rollout** was associated with more strictures than holmium, attributed to higher energy settings and greater tissue energy absorption.
- **Impacted stone >2 months → about one in four develop stricture.**
- **Modern ureteroscopy** (small, actively deflecting, digital) has a long-term complication rate **<5%**. **Access sheaths** caused smooth-muscle-layer injury in up to 13% in one unverified study.

Diagnostic Studies and Indications for Intervention

- **CT** shows obstruction; **antegrade/retrograde pyelography, CT urography or diagnostic ureteroscopy** define location and length.
- **Ureteroscopy with biopsy or barbotage whenever the cause is uncertain.**
- **Diuretic renography before treatment:** endourological therapy generally needs **ipsilateral function $\geq 25\%$** (text; the management algorithm uses **20%**).
- **Indications: exclude malignancy;** ongoing obstruction; **recurrent pyelonephritis;** pain from functional obstruction.

Endourological Options for Intervention

Ureteral Stent Placement

- **Effective acutely for most strictures, especially intrinsic,** with good medium-term success.
- **Extrinsic compression often ends in nephrostomy or surgery.** Chronic stenting here must be “guarded”, since drainage may be short-lived; monitor upper tracts and symptoms.
- **Chronic stenting with periodic changes:** unfit for repair, poor prognosis, or on systemic cancer therapy.

COMPARISON — STENT STRATEGIES FOR CHRONIC URETERAL OBSTRUCTION

Stent	Use	Performance	Problems
Single polymeric stent	Intrinsic stricture; temporising	Good for intrinsic disease	Fails early in extrinsic compression
Tandem stents (two parallel)	Benign or malignant extrinsic obstruction; when a single stent fails	No failures in benign disease; about 1 in 8 fail in malignant disease	—
Full-length metallic stent	Malignant (and selected benign) obstruction	Exchange at 12-month intervals ; up to 3 years' drainage; pooled success about 72.5%, migration about 1%	Exchange difficulty, encrustation; about one-third failed in one malignant series (obstruction, distal migration, subcapsular haematoma)
Expandable metallic mesh	Malignant	—	Encrustation, hyperplastic reaction, tumour ingrowth
Non-mesh thermo-expandable metallic	Malignant	Drainage plus therapeutic benefit	UTI, migration, encrustation, obstruction

- **Pattern:** metallic stents exist to outlast polymeric stents against extrinsic pressure. Their weakness is **tissue ingrowth and encrustation**, and **high-quality evidence is lacking**.

Balloon Dilation

- Rigid dilation was rarely definitive and needed repeating. **Angiographic balloons (early 1980s) + temporary stent** became accepted.
- **Indication:** functionally significant obstruction. **Contraindications:** active infection; stricture >2 cm. **Any endoscopic technique is likely to fail beyond 2 cm.**

Retrograde technique.

- **Sequence:** retrograde pyelogram (site and length) → open-ended catheter to stricture guides **floppy-tipped/hydrophilic wire** across → catheter passed through → exchanged for **high-pressure 4-cm, 5–8-mm balloon** → position by radio-opaque markers → inflate until the **waist disappears** → **10 minutes** → stent **2–4 weeks** → renogram about 1 month after stent removal, then every 6–12 months.
- If the wire will not pass fluoroscopically → **ureteroscopic guidance**, or low-profile balloon through the ureteroscope.

Antegrade technique.

- When retrograde access is impossible. **Nephrostomy first**; with infection or impaired function, drain alone until resolved.
- Antegrade study → wire across → balloon until waist disappears → **internal stent + nephrostomy** → nephrostogram at **24–48 hours** → remove nephrostomy; or use a cappable **internal–external stent**.

COMPARISON — BALLOON DILATION OUTCOME BY STRICTURE TYPE

Favourable	Unfavourable
Iatrogenic, non-anastomotic (e.g. after ureteroscopy): about 85% Short (≤2 cm) and recent (≤3 months) Intact blood supply Short anastomotic transplant strictures on immunosuppression: about 79%	Anastomotic: about 50% Long strictures Poor vascular supply Pooled success falls from about 60% at 3 months to about 54% at 6–12 months

- **Mechanism:** balloon dilation produces **longitudinal splits similar to an endoureterotomy**, which explains its successes.
- No consensus on balloon type, pressure, stent type or duration. Some prefer endoureterotomy first-line in transplant strictures.

Endoureterotomy

- **Logical extension of balloon dilation. Retrograde preferred (less invasive);** antegrade when a tract already exists. Performed **under direct ureteroscopic vision**.
- **Follow with diuretic renography for up to 2 years** to catch late failures.

Retrograde ureteroscopic approach.

- **Sequence:** retrograde study → wire across (fluoroscopically, or through the flexible ureteroscope under vision) → **safety wire always left across** → ureteroscope reintroduced alongside → incise.
- **Tools:** cold knife, cutting electrode, **holmium laser (dominant today)**.
- **Full thickness, lumen to periureteral fat, including 2–3 mm of normal ureter above and below.** Balloon may be used before (to gain access) or after (to enlarge the incision).
- **Stent:** larger (**8–12 Fr**) associated with better results. Ureteroscopic **triamcinolone injection** may help; biological response modifiers may have a future role.

ENDOURETEROTOMY INCISION DIRECTION BY LEVEL

Lower (distal) ureter	Upper (proximal) ureter
Anteromedial (algorithm: anterior for distal and mid-ureter), away from the iliac vessels	Lateral or posterolateral , away from the great vessels

Results.

- **Holmium endoureterotomy 66–83%** (series >10 patients, >12 months). Transplant ureteral strictures succeed in about two-thirds to over four-fifths, often after prior balloon dilation.
- **Retrograde laser endoureterotomy is the attractive first-line option for strictures <2 cm.** Results are poor above 2 cm.

Antegrade approach.

- When retrograde ureteroscopic access fails: nephrostomy, resolve infection/function → dilate tract for a working sheath → flexible ureteroscope → as retrograde. **Safety wire across and coiled in the bladder throughout.**

Combined retrograde and antegrade approach (complete obliteration).

- **Open or robotic reconstruction is best when possible;** the combined approach is for high-risk patients, and **with caution.**
- **Sequence:** simultaneous antegrade + retrograde pyelogram → scopes from both ends, ends localised fluoroscopically → **turn off one light source and cut toward the light** of the opposite scope → recanalise with the **stiff end of a guidewire** (via semirigid scope; or flexible scope/open-ended catheter) or holmium laser → through-and-through wire → **stent 8–10 weeks.**
- **Success inversely related to length.** Internalised drainage, even stent-dependent, is a quality-of-life gain. Cautery-balloon and blind fluoroscopic series report moderate success.

FIG. 93.24 — ALGORITHM FOR MANAGEMENT OF BENIGN URETERAL STRICTURE DISEASE

Assess length and renal function	Branch	Management
Stricture <2 cm and >20% function	Distal ureter	Endoscopic anterior incision
	Middle ureter	Endoscopic anterior incision
	Proximal ureter	Endoscopic lateral incision
Stricture >2 cm	Moiety >20%	Open or laparoscopic repair
	Moiety <20%	Laparoscopic nephrectomy*

- Consider **balloon** if a **transplant on immunosuppression**. ***Children and selected patients with renal insufficiency may warrant repair** despite poor function.
- **What it tests:** two gates, **length (2 cm)** and **function (20%)**, decide endoscopy versus reconstruction versus removal.

KEY POINTS: ENDOUROLOGICAL MANAGEMENT OF URETERAL STRICTURES

- Evaluate to **preserve function and exclude malignancy**; assess function first, since endourology needs **about 25% ipsilateral function.**
- **Chronic stenting for extrinsic obstruction is guarded;** newer stents give long-term success in selected malignant cases.
- Intervene for symptoms or functionally significant obstruction. **Contraindications: active infection; stricture >2 cm.**
- **Retrograde holmium laser endoureterotomy** is an attractive first-line option for short strictures.

- **Lower ureter anteromedial** (iliac vessels); **upper ureter lateral/posterolateral** (great vessels).

Concept — *Endourology succeeds when the stricture is short, recent, non-anastomotic and well vascularised, and the kidney is worth saving. Each failure predictor is a proxy for ischaemic scar that an incision cannot heal across. The 2-cm rule and the function threshold are the two gates.*

Reconstructive Repair

- **Define nature, location and length before any repair:** intravenous pyelogram or antegrade nephrostogram ± retrograde pyelogram. Individualise renography and ureteroscopy/barbotage/brushing to exclude carcinoma.
- **Location and length dictate the operation.**

TABLE 93.2 BRIDGING VARIOUS URETERAL DEFECT LENGTHS WITH DIFFERENT RECONSTRUCTIVE SURGICAL TECHNIQUES

Technique	Defect bridged (Table 93.2)	Value given in the text	Level
Ureteroureterostomy	2–3 cm	“Short defects only”	Upper and mid-ureter
Ureteroneocystostomy	4–5 cm	Distal 3–4 cm of ureter	Distal
Psoas hitch	6–10 cm	Adds up to 5 cm over simple reimplantation	Lower third, below the pelvic brim
Boari flap	12–15 cm	10–15 cm (text and Key Points)	Lower and mid; spiral flap may reach the pelvis (especially right)
Renal descensus	5–8 cm	Up to 8 cm	Upper; adds length to any repair

- **Why the columns differ:** the table gives total defect bridged, whereas the text measures from a different reference (length of distal ureter replaced; length *added* by the hitch). The two are compatible, apart from the **Boari lower limit (10 vs 12 cm)**, which is a true discrepancy.
- **The ladder:** each step adds length. **Transureteroureterostomy (TUU), ileal ureter, buccal graft and autotransplantation** lie beyond it.
- **Trap:** tension-free mobility often can't be judged until surgery, so **be prepared for the next rung.**

Ureteroureterostomy

- **Short upper or mid-ureteral defects** (stricture or recent injury). **Lower ureter** → ureteroneocystostomy ± psoas hitch/Boari flap. Transplant: donor ureteral stricture → anastomosis to healthy native ureter.
- **Tension almost always causes stricture**, so only short defects are suitable.
- **Incision by level:** flank (upper); Gibson or lower midline (mid/lower); reuse a Pfannenstiel for iatrogenic injury, extended in **“hockey-stick”** fashion for proximal access. **Extraperitoneal** unless the injury was transperitoneal.
- **Sequence:** find ureter crossing the iliac vessels → vessel loop/Penrose, **minimal forceps handling, preserve adventitia** → **gunshot: excise devitalised ureter plus a margin of normal-looking ureter** (blast effect) → trim to healthy ends → **spatulate 5–6 mm, 180° apart** (a grossly dilated end is cut obliquely, not spatulated) → corner-to-apex stitches tied outside the lumen → run or interrupt → **double-J before closure** (wire through a mid side-hole stiffens it; **reflux of methylene blue** from bladder confirms distal position) → cover with fat or omentum.
- **Laparoscopic/robotic:** first laparoscopic case was for endometriosis. Paediatric duplex series; robotic series with high 2-year recurrence-free rates and low-grade (Clavien–Dindo ≤2) complications. **Viable for almost any short obstruction.**

- **Postoperative:** drain + Foley 1–2 days. Drain out after 24–48 hours of minimal output. If transperitoneal, **check drain-fluid creatinine**. Stent out at 4–6 weeks.
- **Success >90%. Suspected fistula — sequence:** plain radiograph for stent position → check drain is not on the anastomosis → **stop suction** (straight drainage) → Foley + anticholinergics for reflux/spasm → persistent → **nephrostomy**.

Ureteroneocystostomy

- **Adult injury or obstruction of the distal 3–4 cm;** add psoas hitch or Boari flap to avoid tension.
- **Laparoscopic:** usually transperitoneal (space) with intracorporeal suturing and stent; outcomes comparable with open with less morbidity; LESS reported.
- **Robotic:** 4-arm, prostatectomy-type ports (shifted cephalad). **Same success as open, shorter stay and less opioid;** effective even with hitch or Boari flap.
- **Open vs laparoscopic vs robotic:** similar major complications (10–20%) and failure (6–16%), though **robotic follow-up was much shorter**.

Psoas Hitch

- **Lower-third defects;** adds up to 5 cm. **Defect above the pelvic brim usually needs more than a hitch.**
- **Indications:** distal stricture, injury, failed reimplantation; can combine with TUU.
- **Contraindication:** **small, contracted, immobile bladder.** Urodynamics for capacity/compliance; **treat outlet obstruction or neurogenic dysfunction first.**
- **Success >85% in adults and children.** Rare complications: fistula, obstruction, nerve, bowel or iliac vein injury, urosepsis.

Boari Flap

- When the diseased segment is too long or mobility too limited. **First described in dogs, 1894.** Bridges **10–15 cm;** a **spiralled flap can reach the renal pelvis**, especially on the right.
- **Preoperative:** bladder capacity and function; treat outlet obstruction or neurogenic dysfunction. **Small capacity → consider alternatives.**
- **Results good if well vascularised. Commonest complication: recurrent stricture** (ischaemia or tension). Rare pseudodiverticulum. Proximal strictures succeed with Boari ± downward nephropexy.
- **Renal function after reimplantation** (mostly psoas hitch ± Boari): eGFR fell in 44% within 2 years despite hydronephrosis resolving, **more in men and those with abnormal baseline function.** Useful for counselling.
- Laparoscopic and robotic Boari flaps: transperitoneal in all reports.
- **Refluxing vs non-refluxing reimplantation in adults:** no difference in function or stenosis; effect on pyelonephritis unclear.

COMPARISON — PSOAS HITCH VS BOARI FLAP

Feature	Psoas hitch	Boari flap
Length gained	Up to 5 cm (defect 6–10 cm)	Defect 10–15 cm; can reach pelvis
Complexity	Simpler	Greater
Vascular risk / voiding difficulty	Lower	Higher (flap ischaemia)
Bladder requirement	Mobile bladder; not small/contracted	Adequate capacity
Main complication	Uncommon (fistula, obstruction)	Recurrent stricture

Renal Descensus

- Described in 1964. **Sequence:** enter Gerota's fascia → mobilise kidney fully → rotate **inferiorly and medially on its pedicle** → fix lower pole to retroperitoneal muscle. **Up to 8 cm.**
- **Limited by the renal vein;** dividing and reimplanting it lower on the IVC is possible but rarely done. Combine with Boari flap for **panureteral strictures**; laparoscopic technique reported.

Transureteroureterostomy

- First clinical use 1934. For ureteral length insufficient to reach the bladder.
- **Absolute contraindications:** donor ureter too short to reach the recipient without tension; diseased recipient ureter.
- **Relative contraindications (any process that can affect both ureters):** nephrolithiasis, retroperitoneal fibrosis (RPF), urothelial malignancy, chronic pyelonephritis, abdominopelvic radiation.
- **Recipient reflux must be corrected at the same time** → preoperative voiding cystogram.
- **Sequence:** midline transperitoneal → mobilise colon, then donor ureter with adventitia, and divide just above obstruction → expose only about 5 cm of recipient → **tunnel under sigmoid mesentery above the inferior mesenteric artery (IMA)** (prevents tethering and kinking) → minimal recipient mobilisation → **anteromedial ureterotomy** → spatulated end-to-side anastomosis → double-J from donor pelvis to bladder (± second stent in recipient).
- **Results:** large paediatric series without loss of a single recipient kidney. Revisions were for **IMA kinking**. In complex adults (diversion, radiation) complications occurred in about one-quarter (commonest urine leak) with **patency about 96%** long term. Laparoscopic TUU is feasible without long-term data.

Ileal Ureteral Substitution

- **Long defects, especially proximal. Urothelium-lined tissue is preferred** (non-absorptive, resists urine's inflammatory and carcinogenic effects), so bowel is used only when nothing else bridges or the bladder is unsuitable.
- **Appendix and fallopian tube are unreliable substitutes.** First ileal ureter **1909** (tuberculosis).
- **Physiology:** isoperistaltic ileum to bladder → reflux and pelvic pressure rise **only during voiding**, depending on segment length and voiding pressure. **Segments >15 cm showed no reflux into the pelvis.** Tapering does not alter perfusion pressure or metabolism.
- **Metabolic risk is set by baseline renal function:** normal function → about 1 in 8 have significant problems. **Creatinine >2 mg/dL → nearly half develop hyperchloraemic metabolic acidosis** needing conversion to a conduit. Bladder dysfunction adds complications.
- No proven advantage of tapering, a non-refluxing anastomosis or segmental replacement.
- **Contraindications:** creatinine >2 mg/dL; bladder dysfunction or outlet obstruction; inflammatory bowel disease; radiation enteritis.
- **Sequence:** bowel preparation (often) → long midline → mobilise colon, dissect ureter to healthy tissue (pelvis if whole upper ureter diseased) → measure defect → distal ileum **≥15 cm from the ileocaecal valve**, confirm blood supply, divide mesentery generously → mark distal end, restore continuity → deliver through colonic mesenteric window (or reflect caecum/ascending colon on the right) → **check isoperistalsis** → anastomose to pelvis or lower-pole calyx and bladder.
- **Ileocalycostomy** for a scarred or intrarenal pelvis, **resecting lower-pole parenchyma** as in ureterocalicostomy. Colon may substitute if more accessible. **Bilateral:** one long segment kidney-to-kidney-to-bladder, or two segments.

- **Complications:** early leak or urinoma; obstruction (oedema, **mucus plug**, kink); **ischaemic necrosis** (suspect if acute abdomen). **Progressive dilation + worsening metabolic state → look for bladder outlet obstruction.**
- **Malignancy in ileal ureter** is reported rarely → **endoscopic surveillance from year 3.** Larger long-term series found none.
- **Long-term:** most ileal ureters remain in situ, and creatinine is stable or improved in the great majority. Acidosis needing treatment affects about one in five.
- **Refluxing anastomosis is safe into a functional bladder. Non-refluxing** is useful where the reservoir will be **colonised** (continent cutaneous diversion), to cut pyelonephritis risk.

Minimally Invasive Ileal Ureteral Substitution

- Laparoscopic (transperitoneal three-port) and robotic series: **less analgesia and faster convalescence** than open, with short follow-up.
- Complications in fully intracorporeal robotic series include **volvulus with segment ischaemia.** Longer follow-up needed.

Buccal Mucosa Grafting

- Borrowed from urethroplasty; first human report 1999. Usually an **onlay**, open or minimally invasive.
- **Robotic onlay with omental flap wrap** for blood supply: most patients stricture-free at about 2 years; median stricture length about 3–4 cm; about half redo cases.
- **CORRUS** (Collaborative of Reconstructive Robotic Ureteral Surgery): recurrent strictures after failed prior repair. Buccal onlay at UPJ, proximal and mid-ureter, with failure roughly 15–25% by site.
- **Best niche:** ureter encased in dense scar, where circumferential dissection risks injury or devascularisation.

Autotransplantation

- First performed **1963** for proximal ureteral injury.
- **Consider when the contralateral kidney is absent or poor, or other repairs are not feasible.**
- Harvest as for live donation (maximal vessel length) → iliac vascular anastomoses → **proximal ureter or renal pelvis to bladder.**
- **Laparoscopic harvest** (usually transperitoneal; retroperitoneal possible) + Gibson-incision implantation → less analgesia, faster recovery.
- Success about 90% long term; in benign series (many for loin pain haematuria syndrome) about one-fifth had Clavien–Dindo ≥3a complications, with occasional graft loss.

KEY POINTS: SURGICAL REPAIR OF URETERAL STRICTURE

- **Only short defects** suit end-to-end ureteroureterostomy.
- **Distal stricture → ureteroneocystostomy** ± psoas hitch or Boari flap.
- **Boari flap bridges 10–15 cm;** small bladder capacity contraindicates it; ensure flap blood supply.
- **TUU contraindicated** by disease that may affect both ureters or by a donor ureter too short to reach tension-free.
- **Ileal ureter** for extensive loss; contraindicated by **creatinine >2 mg/dL**, bladder dysfunction or outlet obstruction, inflammatory bowel disease, or radiation injury (enteritis).
- **Autotransplantation** when the contralateral kidney is absent or poor, or other repairs are not feasible.

- Laparoscopic or robotic reconstruction may speed recovery in skilled hands.

Concept — Ureteral reconstruction is a **length ladder**. Bring the bladder up (hitch, Boari flap), bring the kidney down (descensus, autotransplant), borrow the other ureter (TUU), or substitute tissue (ileum, buccal mucosa). Each rung trades more length for more morbidity. **Tension and ischaemia cause failure at every level.**

URETEROENTERIC ANASTOMOTIC STRICTURE

Incidence and Aetiology

- **Conduits: stricture 4–9%, more common on the left. Continent diversions: 3–25%, mostly within 2 years;** about 5% of strictures appear after 10 years, so **long-term upper-tract surveillance** is needed.
- **Ureteral ischaemia is central.** Blood supply runs **in the adventitia**, so stripping it causes stricture. Mobilise enough for a tension-free join, no more.
- **Why the left:** it needs extra length and dissection to pass **under the sigmoid mesentery over the aorta**, and may angulate around the **IMA**.
- **Most strictures are benign;** carcinoma in situ at the ureteral margin is a risk factor.
- **Technique modifiers:** shield-shaped rather than slit ileotomy lowers stricture; **intraluminal visualisation** (conduit opened on antimesenteric border) is associated with low rates; **routine stenting** has advantages; **indocyanine green fluorescence** in robotic cystectomy may reduce benign strictures.
- **Other risk factors:** higher body mass index; **American Society of Anesthesiologists (ASA) score >2**; nodal involvement; Clavien–Dindo grade 3–4 complications within 30 days; **prior abdominal surgery**; **male sex**. Leads to ultrasound upper-tract surveillance for the first decade.

COMPARISON — ANASTOMOTIC CHOICES AND STRICTURE RISK

Comparison	Finding	Verdict
Non-refluxing vs refluxing ureteroenteric anastomosis (including a randomised trial into an afferent ileal limb)	Non-refluxing ≈13% vs refluxing ≈2–3% stricture; >5-fold increase in one series, independent of surgical expertise ; no difference in hydronephrosis, pyelonephritis, stones or renal insufficiency	Refluxing anastomosis into low-pressure reservoirs: reflux into an adult kidney is not clearly harmful, but obstruction is
Colon vs ileum conduit	Colon permits non-refluxing anastomosis; no clear renal benefit	Controversial
Wallace (joined ureters) vs Bricker (separate)	Both low; Wallace none , Bricker about 3.7% (Bricker group heavier)	Both acceptable
Open vs robotic cystectomy	Conflicting: no difference in one; robotic higher in others (intracorporeal highest early, falling with experience)	Learning curve matters

Evaluation

- **Screening after any diversion:** ultrasonography, CT or magnetic resonance imaging (MRI); CT or MRI if stone or recurrent tumour suspected.
- **Evaluate if** renal colic, recurrent UTI or loss of function.
- **Hydronephrosis:** CT urography, **loopogram** or antegrade nephrostogram (length and site) + **diuretic renography** (function; confirms functional obstruction).

- **Function too poor for urography or renography → nephrostomy + antegrade nephrostogram:** diagnostic, therapeutic, and defines length for planning.

Initial Management and Intervention

- **Not all hydronephrosis in a diversion is obstruction.** Long-term conduits commonly show chronic non-obstructive dilation.
- **Triggers for renography:** falling function or **loss of reflux on routine loopogram.**
- **Intervene for** pain, infection or renal insufficiency with functional obstruction.
- **Irregular mass at the stricture + rapid progression → suspect recurrent transitional cell carcinoma.**
- **Pelvic exenteration (mostly irradiated):** high early and late anastomotic complication rates, but most are managed conservatively (e.g. nephrostomy).
- **Open revision remains the gold standard** (about two-thirds success vs about one-quarter endoscopic), though it carries about one-third complications. **Robotic repair** is equally successful with half the stay. **Ureteroileal bypass** is a novel option needing less ureteric dissection.

Endourological Management

- Evolved from balloon plus stent → electrocautery incision → **small flexible ureteroscope + holmium laser** (current standard).
- **Antegrade favoured** (unlike ordinary ureteral strictures) because a **nephrostomy is often already present.**
- **Sequence:** continue nephrostomy until stable → antegrade wire across → balloon until waist disappears (or incise) → stent antegrade. **Internal-external stent** preferred because of **mucus plugging** (flushable, changeable over wire). Optional retrograde looposcopic access for through-and-through control and direct vision → **stent 4–8 weeks.**

COMPARISON — ENDOUROLOGICAL VS OPEN TREATMENT OF URETEROENTERIC STRICTURE

Modality	Durable success	Note
Balloon dilation	Short-term up to about 60%; long-term poor (5–30%); repeat dilation often needed	First endourological method
Metallic stents	Patency >80% at 6–22 months	More encrustation, stone, ingrowth, migration
Antegrade laser endoureterotomy	50–80%; intraluminal invagination about 80% (side-independent); laser + triamcinolone about 83%	Current endoscopic tool; direct vision favoured over fluoroscopic
Endourology overall vs open	About 26% vs 91%	—
By length: <1 cm vs >1 cm	About 50% (up to 100% in small series) vs 6%	Length is decisive
By side: right vs left (endourology)	About 41% vs 19%	No side effect with open repair

- **Left-sided danger:** incision risks **haemorrhage from the adjacent sigmoid mesentery.** With poorer left-sided success, **consider primary formal repair for left strictures.**

FIG. 93.29 — ALGORITHM FOR MANAGEMENT OF URETEROENTERIC ANASTOMOTIC STRICTURE DISEASE

Assess length and renal function	Branch	Management
Stricture <2 cm and >20% function	Right	Endoscopic incision
	Left	Consider formal repair

Assess length and renal function	Branch	Management
Stricture >2 cm	Moiety >20%	Consider formal repair
	Moiety <20%	Nephrectomy or observation*

- Consider balloon if transplant on immunosuppression. *Children and selected patients with renal insufficiency may warrant repair.
- **Contrast with Fig. 93.24:** the same length and function gates, plus a **third gate, side. Left short strictures go to formal repair**, and **observation** is an option for a poor moiety.

KEY POINTS: URETEROENTERIC STRICTURES

- Minimally invasive long-term patency is only **about 50%**, yet it remains first-line, with surgery for failures.
- **Left conduit strictures:** sigmoid mesentery haemorrhage risk and low endoscopic success → **definitive repair**.
- Open/robotic repair gives acceptable long-term success. **>1 cm and left side predict recurrence**.

Concept — A ureteroenteric stricture is an **ischaemic scar at a surgically devascularised ureter**. That is why the left (longer, tunnelled, angulated) and the non-refluxing tunnel (more dissection) stricture more, and why an incision, which cannot restore blood supply, fails on long or left-sided strictures.

RETROPERITONEAL FIBROSIS

Presentation and Aetiology

- Rare inflammatory fibrotic process compressing retroperitoneal structures, including the ureters.
- **Age 40–60** (over 30 paediatric cases reported); **male:female 2–3:1** (one recent series found no sex difference); incidence about **1 per 200,000–500,000 per year**.
- **Mass centres on the distal aorta at L4–L5** and wraps the ureters → hydronephrosis by **extrinsic compression or interference with peristalsis**.
- **Symptoms: lower back/flank pain (commonest)**; weight loss, anorexia, nausea, malaise, fever, hypertension, oliguria/anuria. **Duration 4–6 months** before diagnosis; about half present with significant obstruction and **uraemic** symptoms.
- **Extension:** IVC → deep venous thrombosis and leg oedema. Renal hilum and vein → renal venous hypertension and **gross haematuria**. Iliac bifurcation, spermatic cord and scrotum. Aortic obstruction and mediastinal, biliary, mesenteric or renal involvement are rare.

IDIOPATHIC VS SECONDARY RPF

Idiopathic (about 70%)	Secondary (about 30%)
Part of chronic periaortitis, a large-vessel vasculitis Proposed antigen: ceroid (oxidised lipid-protein polymer in atherosclerotic plaque); more aortic aneurysms May be part of multifocal fibrosclerosis: sclerosing mediastinitis, sclerosing cholangitis, orbital pseudotumour, Riedel thyroiditis Probably autoimmune: associations with membranous glomerulonephritis, ankylosing spondylitis, granulomatosis with polyangiitis (Wegener); antinuclear antibody positivity predicts relapse	Drugs (commonest identifiable cause): methysergide and other ergot alkaloids; also β-blockers, phenacetin Malignancy: lymphoma (commonest), myeloma, carcinoid, pancreas, prostate, sarcoma Radiotherapy for retroperitoneal malignancy Asbestos exposure Infection: tuberculosis, <i>Actinomyces</i>, gonorrhoea, schistosomiasis

- **Pathology:** smooth, flat, tan, dense mass; may **invade ureter or psoas**.
- **Histology by stage:** early = collagen + capillary proliferation + lymphocytes, plasma cells, fibroblasts. Late = **acellular, avascular sheets of hypocellular collagen**.
- **Trap:** malignant RPF is **histologically indistinguishable** except for small islands of tumour cells, so **deep and multiple biopsies** are needed.

Evaluation

- **Symptoms non-specific; examination usually unrevealing.**
- **Laboratory:** raised erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) in one-half to two-thirds; moderate leucocytosis; anaemia; variable renal insufficiency with electrolyte disturbance.
- **CT urography (if function normal): hydronephrosis + medial deviation of proximal and mid-ureter + smooth tapering** at obstruction. **Usually bilateral**; some have symptoms with little hydronephrosis.
- **CT:** well-delineated homogeneous soft-tissue mass enveloping great vessels and ureters. **Significant renal impairment → retrograde pyelography.**
- **MRI:** low T1 signal throughout; T2 variable, and high T2 signal = **active disease**. T2 signal and gadolinium enhancement **fall with effective therapy** (visible at 1 month).
- **Positron emission tomography (PET) is the most sensitive test of disease activity** and may obviate biopsy (100% negative predictive value in one series). PET-CT is used for follow-up.
- **Suspected non-functioning kidney → differential renography** (planning). **Biopsy** percutaneously or at surgery if diagnosis remains uncertain.

Management**Initial Management**

- **Hydronephrosis + uraemia → emergent decompression** by nephrostomy or stents. **Critically ill with electrolyte derangement and little or no urine → nephrostomy favoured.**
- **Then watch for post-obstructive diuresis;** replace fluid and electrolytes.
- **Then find the cause: stop methysergide or any culprit drug;** search for occult malignancy; biopsy percutaneously or at ureterolysis.
- **Biopsy may be omitted before medical therapy** if there is no malignancy history, no atypical MRI/CT features and no lymphadenopathy, a position strengthened by PET.

Medical Management

COMPARISON — MEDICAL THERAPY FOR IDIOPATHIC RPF

Agent	Line	Regimen / role	Note
Corticosteroids (prednisone)	First line	60 mg daily tapered to 5 mg; up to 2 years	About 80% response; relapse on taper 25–50%
Azathioprine, cyclophosphamide, ciclosporin, mycophenolate mofetil, colchicine	Second line (steroid failure)	Immunosuppression	Isolated reports
Tamoxifen; medroxyprogesterone, progesterone	Alternative / second line	Thought to inhibit fibroblast proliferation	Tamoxifen particularly beneficial
Rituximab	Alternative / second line	Biological agent	Reported benefit

- **Steroid response is rapid:** pain and constitutional symptoms resolve within days, ESR falls fast, diuresis follows; the mass shrinks and ureteral/IVC compression improves.
- **Active inflammation predicts response:** raised ESR, CRP, leucocytosis, or active inflammation on biopsy.
- **Drainage then medical therapy** resolves obstruction in about two-thirds, with low recurrence; only about 15% need ureterolysis.

Surgical Management — Ureterolysis

- Open historically (**midline transperitoneal** for both ureters); laparoscopic/robotic in modified flank position, turning for the second side.
- **Bilateral ureterolysis even if hydronephrosis is unilateral**, since the process is usually bilateral.
- **Sequence: stents/catheters first** (identification) → mobilise colon → **deep biopsies for frozen and permanent section** → **start at the distal, non-dilated ureter** (saves the thin dilated segment) → “**split and roll**”: right-angle clamp between ureter and mass, incise fibrosis over it, repeat → close any inadvertent ureterotomy with absorbable suture (**excision and ureteroureterostomy rarely needed**).
- **Then separate the freed ureters from the fibrotic bed**, or it recurs. Use healthy, well-vascularised tissue and **avoid kinking at the peritoneal hiatus**.

COMPARISON — URETERAL PROTECTION AFTER URETEROLYSIS

Method	How	Note
Intraperitonealisation	Ureters lateralised; parietal peritoneum closed over the bed (or reflected colon placed behind the ureter, edge fixed to side wall)	Radiological and clinical outcomes equal to lateral retroperitoneal placement
Lateral retroperitoneal placement	Ureter displaced laterally	Equivalent to intraperitonealisation
Omental wrap	Omentum freed from transverse colon → split in midline → short gastrics divided → two halves on right and left gastroepiploic arteries → wrap whole ureter	Most definitive for extensive RPF: barrier plus blood supply to an ischaemic ureter

- **Postoperative steroids** may prevent recurrence. **Stents removed early** if no ureterotomy.
- **Ureterolysis impossible** → **autotransplantation** if function is satisfactory. **Unrecoverable function after adequate decompression + good contralateral kidney** → **nephrectomy**.

Outcomes of Ureterolysis

- **Open:** about 94% stent-free at 12 months. Ureteral reconstruction or ileal ureter is also successful when needed.
- **Laparoscopic (first 1992):** occasional conversion (iliac vein injury, dense fibrosis); long operation (bilateral much longer than unilateral); **obstruction-free in about 90%** at about 2.5 years; multi-centre success about 83%. Tends to be done where a fellowship-trained laparoscopist is available.
- **Laparoscopic vs open:** shorter stay with equal success and complications.
- **Robotic ureterolysis + omental wrap:** about 86% patent at about 20 months. One enterocutaneous fistula came from unrecognised thermal bowel injury.

KEY POINTS: RETROPERITONEAL FIBROSIS

- **Mass centres on the distal aorta at L4–L5** and wraps the ureters (compression or impaired peristalsis). **Mostly idiopathic**, associated with chronic periaortitis.
- Non-specific features; **ESR, CRP**, leucocytosis, anaemia, variable renal insufficiency.
- **Hydronephrosis + uraemia → emergent nephrostomy or stents**, then watch for **post-obstructive diuresis**.
- **Steroids first line**; immunosuppressants second.
- **Bilateral ureterolysis with intraperitonealisation or omental wrap**; open, laparoscopic or robotic; autotransplantation possible.

Concept — *RPF is an inflammatory periaortitis that obstructs by encasement. Management follows the disease activity: decompress, exclude secondary causes, suppress the active inflammation (steroids work while it is cellular), and operate on the burnt-out, acellular fibrosis, isolating the ureter so it cannot be re-encased.*